GitHub Repository link

https://github.com/ExplosiveSalad/Data-Process-and-Software-Modelling.git

Task 1: Organisational and System Analysis

- **Overview of the selected organisation**

Waitakere Hospital is a public hospital, operating as a facility under Te Whatu Ora - Health New Zealand Waitemata (formerly Waitemata District Health Board). Health New Zealand delivers publicly funded health services to approximately 650,000 residents across the North Shore, Waitakere, and Rodney regions. Following the redevelopment funding approved in 2000, the hospital was refurbished and reopened in 2005 with expanded facilities, those being medical beds, coronary care beds, an Emergency Department, a Surgical Unit, elective surgical and endoscopy services, a Maternity Unit, acute gynaecology, elective gynaecology, and specialist inpatient wards.

- **Departmental structure and responsibilities**

The hospital is structured around acute, inpatient, and community-focused services. These departments all interact with the Patient Admission, Triage and Bed Management System (PATBMS).

The Emergency Department (ECC-Emergency Care Centre): A 24-hour facility responsible for treating acute illness and injuries, including specialised segregated areas for adult and paediatric patients.

The Assessment and Diagnostic Unit (ADU): A short-stay unit that acts as a gateway between the ECC and inpatient wards. The Surgical Unit features operating theatres, including a dedicated obstetric (pregnancy and childbirth) theatre, and an endoscopy room for elective and acute surgery.

The Mental Health Unit: An inpatient mental health unit located on site with community services based nearby.

The Allied Health Team: A team of social workers, physiotherapists, occupational therapists, and speech language therapists that work alongside Nurses and Doctors to provide the proper treatment for patients. The Medical Wards are general adult medical services with specific named wards such as Te Henga, Piha, Titirangi and a few more.

- **Identification of stakeholders and user roles**

The stakeholders at Waitakere Hospital are involved in the delivery, management, and consumption of health services. The key stakeholders here are the patients and families. These are the primary recipients of care, including those utilizing specialized services like the Children's Ward, maternity, and emergency services. The clinical staff, such as senior medical staff (doctors and specialists), nurses, midwives, allied health professionals, and pharmacists. The Hospital Leadership and Governance include the Executive Governance Group, project managers, and the Staff and Consumer Advisory Group. The Service Support and Administrative Staff consist of dedicated staff managing patient transport, hospital security, cleaning, and catering. The Regional Health Partners are the owner of this facility, Te Whatu Ora (Health New Zealand), and others like PHOs (Primary Health Organisations), regional Public Health Units, and GP Liaisons. The Community and Advocacy Groups which are the disability advocates, Māori, Pacific, and Asian Health Groups, as well as community organisations. Finally, the last stakeholders are the Support Partners which are The Well Foundation who fundraise money for items like the Waitakere Hospital ICU.

The PATBMS is used daily by these five key user roles, each with different responsibilities and access levels.

- **Doctors/Specialists:** They are responsible for entering diagnoses, treatment plans, and clinical decisions; they review triage outcomes and authorise patient admissions or transfers.
- **Nurses/Midwives (including charge nurses):** They provide direct patient care, update real-time status in the system, perform initial triage scoring, and manage ward-level bed allocation and handovers.
- **Allied Health Professionals:** Record assessment results, therapy plans, and send out information to support multidisciplinary care coordination.
- **Administrative and Health Information Staff:** They are responsible for registering patients, searching for patient records via NHI number and processing clinical decisions and documentation from doctors.
- **Hospital Management/Executives:** Monitor overall bed occupancy, generate reports on patient inflow and capacity, and authorise overrides for urgent cases that affect regional service planning.

- **System scope, boundaries, and assumptions**

The system scope of Waitakere hospital covers core services, which are 24/7 acute emergency medicine, inpatient services, maternity services, elective surgery, and rehabilitation. Only patients that meet a specific clinical requirement are admitted for elective surgery, otherwise more pressing or urgent cases usually override these rules. Specialist areas include paediatric and child health, women's health, orthopaedics, and general medicine and mental health services such as the mason clinic and forensic psychiatry. Patients are usually triaged in the Emergency Department and ordered by how severe their symptoms are and how available they currently are, to which they are then admitted to the relevant ward or transferred to specialist units. The infrastructure has the Totara Haumaru building, with fully equipped theatres and high-dependency beds. Finally, the support services are a multidisciplinary team care that include nursing, allied health, and cultural support services like Māori health plans and Pacific navigators. Geographically, it is bounded to the west Auckland area and mainly that only. It operates within the Waitemata region but refers high level tertiary cases (e.g. complex cardiac surgery) to Auckland Hospital. The hospital is a secured facility with access control. It has limited entry after 8pm, relying on security personnel for entry. Waitakere Hospital operates under the national Health New Zealand system and adheres to the Code of Health and Disability Services Consumers Rights. The assumptions I am going to make are that the system is rapidly growing, with a diverse population with a high percentage of it being young people and specific cultural needs, more so requiring better focused health planning. Another thing is the safety protocols; I am assuming that it has 24/7 security to maintain a zero-tolerance violence policy towards staff. A technological assumption I am making is that the system uses real-time data to make accurate clinical decisions and proper patient management. The hospital system delivers real time data between wards, diagnostic services, and surgical theatres to ensure they are proficient in treating those who need it and allocating beds in time.

Task 2: Requirements Engineering

Under this is the functional and non-functional requirements for the PATBMS which was derived from the stakeholders, user roles, and system scope mentioned in Task 1.

- **Clearly defined functional requirements**

Before going into functional requirements, I will quickly recap what a functional requirement is. Functional requirements describe the system-specific behaviours that must be implemented.

FR1: The PATBMS (Patient Admission, Triage and Bed Management System) must allow doctors to enter patient diagnoses and treatment plans.

FR2: The PATBMS must allow doctors to review triage outcomes and authorise patient admissions or transfers.

FR3: The PATBMS must enable doctors or Administrative Staff to search for patients via their National Health Index (NHI) number to avoid confusion caused by duplicate names.

FR4: The PATBMS must provide users with a real time bed availability dashboard that visually displays hospital wards and individual bed status using a colour coded map.

FR5: The PATBMS must notify users when no suitable bed is available for a patient and must allow them to request a ward or hospital patient transfer, including the clinical urgency rules.

FR6: The PATBMS must send approved diagnoses, treatment plans, and transfer authorisations from doctors to the Health Information/Administrative staff for further processing and deliberation.

FR7: The PATBMS must take patients' information and NHI number to link it to an already existing file or create a new one upon their arrival.

FR8: The PATBMS must allow nurses and midwives to check if they can triage patients by assigning an Australasian Triage scale (ATS) category. (ATS 1 = Immediately life threatening, ATS 2 = Imminently life threatening, ATS 3 = potentially life threatening, ATS 4 = Potentially serious, ATS 5 = less urgent)

FR9: The PATBMS must have automatic bed allocation or suggest an allocation according to priority, severity, ward specialty, and other clinical factors of the patient. The suggested allocation must require approval by a doctor or charge nurse before finalisation.

FR10: The PATBMS must support structured inter-departmental handovers by allowing the sending department (such as ECC or ADU) to list crucial handover information (such as clinical status, outstanding tasks, special needs) and notify the receiving department (such as the Surgical unit or medical ward).

FR11: The PATBMS must record patient discharges and instantaneously update the associated bed status once discharge has been confirmed.

FR12: The PATBMS must generate and display department-level capacity alerts (bed occupancy) that are accessible to hospital management and charge nurses, with the ability to notify relevant personnel.

FR13: The PATBMS must require users to log in with verified credentials.

- **Clearly defined non-functional requirements**

NFR1: The PATBMS must have a fast response time relating to the bed searches and dashboard updates (The target is less than 2 seconds for standard loads) and must handle large loads of data during unforeseen high-demand events such as multi-casualty events.

NFR2: The PATBMS must maintain 24/7 access so that authorised users are always able to access information and the system must also have real-time data updates of patient and bed status. In the event of a system failure, users can revert to paper records or access an external hard drive for information.

NFR3: The PATBMS must implement role-based access control based on a user's access level; they are allowed to view and edit certain information that is consistent with their access level. All changes will be logged with a history log that allows for viewing and reversion. Patient data must be protected in compliance with the Health Information Privacy Code of 2020, that includes encryption methods and access safeguards.

NFR4: The PATBMS must have an easy-to-use interface for clinical staff working under time pressure or a high inpatient load. The interface must support cultural safety requirements for Māori, Pacific, and other culturally diverse patients and Whanau.

NFR5: The PATBMS must be easy to expand on in response to the growing and diverse West Auckland population which includes the ability to handle large patient volumes without the degradation of the system itself.

NFR6: The PATBMS must send alerts every 30 minutes to the relative ward staff if the bed status data hasn't been updated to keep the information real time. If the system is temporarily unavailable, the system must be able to continue in a limited mode, send notifications to users, and fallback to paper based documentation.

NFR7: The PATBMS must be able to integrate with the National Health Index and must adhere to the Code of Health and Disability Services Consumers Rights.

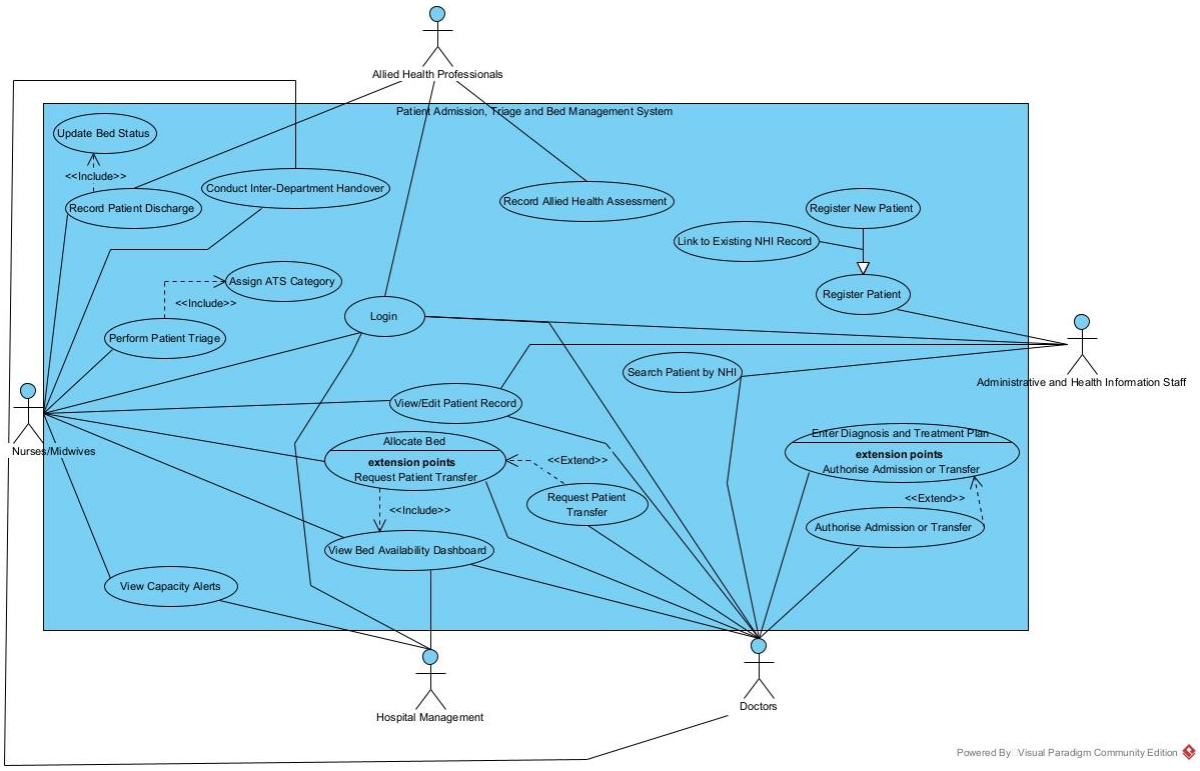
- **Traceability between stakeholders and requirements**

User Role	Related FRs	Related NFRs	Departments Involved
Doctors/Specialists	FR1, FR2, FR3, FR4, FR5, FR6, FR9, FR10, FR13	NFR1, NFR3, NFR4	ECC, ADU, All wards
Nurses/Midwives and Charge Nurses	FR4, FR8, FR9, FR10, FR11, FR12, FR13	NFR1, NFR4, NFR6	ECC, ADU, Surgical Unit, Medical Wards,
Administrative Staff	FR3, FR6, FR7, FR13	NFR3, NFR7	All departments
Allied Health Professionals	FR9, FR11, FR13	NFR4	Medical Wards, Surgical Unit
Hospital Management	FR4, FR12, FR13	NFR1, NFR2, NFR5	All departments

These requirements form the basis for the UML modelling in Task 3.

Task 3: UML Requirements Modelling

The Use Case Diagram below displays the full scope of the PATBMS across all five user roles and key departments. The diagram represents proper use of Include, Extend, and generalisation as per the requirements. All relationships are traceable back to the traceability table back in Task 2.


Allied Health Professionals
Patient Admission, Triage and Bed Management System
Update Bed Status
<<Include>>
Record Patient Discharge
Conduct Inter-Department Handover
Record Allied Health Assessment
Register New Patient
Link to Existing NHI Record
Register Patient
Assign ATS Category
<<Include>>
Perform Patient Triage
Login
Administrative and Health Information Staff
Search Patient by NHI
View/Edit Patient Record
Nurses/Midwives
Allocate Bed
extension points
Request Patient Transfer
<<Extend>>
Request Patient Transfer
<<Include>>
View Bed Availability Dashboard
Enter Diagnosis and Treatment Plan
extension points
Authorise Admission or Transfer
<<Extend>>
Authorise Admission or Transfer
View Capacity Alerts
Hospital Management
Doctors
Powered By Visual Paradigm Community Edition

- **Minimum of five fully dressed Use Case Descriptions**

ID	**UC1**
Name	Perform Patient Triage
Description	Nurse assesses a patient and assigns an ATS category based on urgency
Pre-condition	Patient must be registered in the system
Pre-condition	Nurse is logged in
Event flow	1. Nurse opens triage screen 2. Nurse selects registered patient 3. Nurse enters patient vitals and symptoms 4. Include “Assign ATS Category” 5. Nurse accepts or changes ATS category 6. System records the assigned ATS category to the patient record
Post-condition	Patient has an assigned ATS category in the system
Extension points	None
Triggers	Patient arrives at the Emergency Department

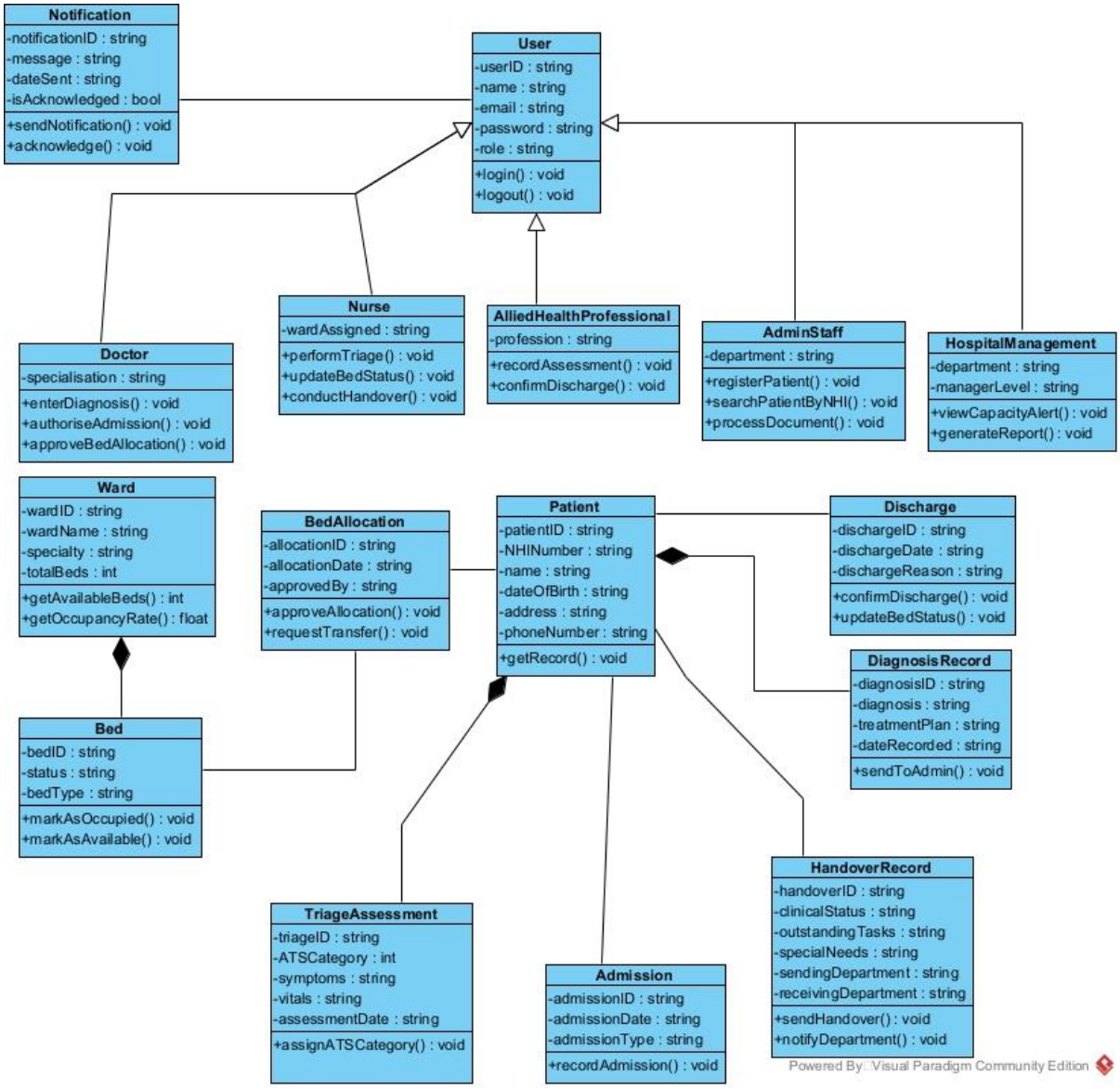
ID	**UC2**
Name	Allocate Bed
Description	Nurse or doctor allocates a bed to a patient based on priority and ward availability
Pre-condition	Patient must be registered in the system
Pre-condition	Patient must have an allocated ATS category
Pre-condition	There must be beds available
Event flow	1. Include “View Bed Availability Dashboard” 2. Nurse or doctor checks for available beds by ward and specialty 3. System suggests a bed based on the priority, severity, and ward specialty 4. Doctor or charge nurse review and approve the suggested allocation 5. System assigns the bed to the patient and updates the bed status
Post-condition	Bed is now allocated to the patient and is marked as unavailable on the dashboard
Extension points	Request Patient Transfer extends Allocate Bed - if no suitable bed is available
Triggers	Patient needs admission after triage or assessment

ID	**UC3**
Name	Register Patient
Description	As soon as a patient walks in, NHI number and information is taken from the patient and either a new patient is registered, or the pre-existing folder is shown
Pre-condition	Administrative Staff is logged in
Event flow	1. Patient provides NHI number and personal information 2. Administrative Staff enters details into the system 3. System checks for an existing folder using the NHI number 4. If pre-existing folder does exist, the system retrieves it and links it, “Link to Existing NHI Record” 5. If pre-existing folder does not exist, the system creates a new patient folder, “Register new Patient” 6. System confirms the registration and makes the patient record available to clinical staff
Post-condition	Patient is registered in the system, and their record is accessible to the relevant and authorised users.
Extension points	None
Triggers	Patient walks into the hospital

ID	**UC4**
Name	Conduct Inter-Department Handover
Description	Nurses or doctors can use the system to send structured handover information to the receiving department that makes sure the patient can be cared for
Pre-condition	Patient must be registered in the system
Pre-condition	Patient must have an ATS category in the system
Pre-condition	Nurse or doctor is logged in
Event flow	1. Nurse or doctor opens the handover screen 2. Nurse or doctor selects the patient record 3. Nurse or doctor enters the patient's status, tasks needing to be done, and special needs 4. Nurse or doctor selects the receiving department 5. The system sends the handover information to the receiving department 6. Receiving department is notified of the incoming patient and the handover details
Post-condition	Handover information is recorded in the system, and the receiving department has been notified
Extension points	None
Triggers	Patient requires transfer to another department for more specialised treatment

ID	**UC5**
Name	Enter Diagnosis and Treatment Plan
Description	The doctor enters the patient's diagnosis and treatment plan into the system after a clinical assessment
Pre-condition	Doctor must be logged in
Pre-condition	Patient must be registered in the system
Event flow	1. Doctor searches for the patient via NHI number 2. Doctor opens the patient's folder 3. Doctor records the diagnosis and treatment plan into the system 4. The system records the diagnosis and treatment plan and assigns it to the patients' folder 5. System sends the approved diagnosis and treatment plan to administrative staff for processing
Post-condition	Patient has a diagnosis and treatment plan on their record in the system and Administrative Staff have been notified
Extension points	Authorise Admission or transfer, If the treatment plan requires a patient's admission or transfer
Triggers	Patient needs a formal diagnosis and treatment plan after a clinical assessment

Task 4: Initial Structural Design and Implementation Planning



- **Mapping between UML Classes and Planned C# classes**

Each UML class corresponds to a C# class. UML attributes correspond to C# private fields shown through public properties, and UML operations correspond to public C# methods. The User class is implemented as a public base class in C# and it has shared attributes such as userID, name, email, password, and role as private fields. login() and logout() are implemented as public operations. Doctor, Nurse, AlliedHealthProfessional, AdminStaff, HospitalManagement all extend User via inheritance, where they all inherit User properties and operations whilst adding their own attributes and operations. The Patient class is implemented as a public class of its own. TriageAssessment and DiagnosisRecord are both implemented as separate classes which are instantiated within Patient, showing the composition relationship displayed in the UML diagram; if a patient object is removed, the TriageAssessment

and DiagnosisRecord objects are also removed. The Ward class contains a list of Bed objects, to which are implemented in C# as a List of Bed, showing the composition relationship where Bed cannot exist without Ward. Bed Allocation, Admission, Discharge, and HandoverRecord are implemented as individual classes that contain references to Patient and Bed objects through association. The Notification class is implemented as an individual class that holds a reference to a User object, allowing Notifications to be sent to any user role in the system. The notification class uses a Boolean isAcknowledged field to track whether a user has seen and responded to the notification. This reflects NFR6 which requires the system to notify ward staff when the bed status has not been updated. The Boolean allows the system to distinguish between sent and acknowledged notifications, supporting real time data across the system.

- **Identification of anticipated design weaknesses prior to OOP application**

Currently the design implements no sort of interface. Proper OOP concepts call for behaviours such as sending notifications or generating reports should be defined through interfaces so that classes aren't tightly coupled, following the principle of high cohesion, low coupling. The role attribute in User is a string meaning that nothing stops someone from entering an invalid role. Proper OOP concepts call for an enum or possibly enforcing roles through the class hierarchy itself. Some classes have limited separation of responsibilities meaning that some are doing too much, such as Discharge where it has an updateBedStatus() method but updating bed status isn't the responsibility of Discharge but Bed class. This overlap shows the design weakness. There is no explicit access control enforcement, FR13 and NFR3 call for role-based access control but the class diagram has no sort of mechanism that enforces which user roles can access which operation. The design assumes that this is handled externally rather than being built into the classes. These weaknesses will be addressed later down the line in a subsequent design phase through the application of object-oriented principles including interfaces, encapsulation, and access control mechanisms. Resolving these issues such as implementing notifications will improve the overall maintainability, security, and scalability of the PATBMS.